

## PIURA, Peru

WMO: 844010

Lat: 5.206S

Lon: 80.616W

Elev: 35

StdP: 100.90

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 8             | 15.9       | 16.4 | 12.9                          | 9.3 | 16.7 | 13.5 | 9.7 | 17.3 | 8.8                   | 23.7 | 8.2 | 23.1 | 1.4                   | 150  | 0.328 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 9.3                    | 34.1            | 25.3 | 33.3 | 24.9 | 32.7 | 24.7 | 26.3                | 32.2 | 25.9 | 31.8 | 25.4 | 31.2 | 4.1                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.8                            | 19.9 | 29.4 | 24.2 | 19.2 | 29.2 | 23.6 | 18.5 | 29.3 | 82.1          | 32.1 | 80.5 | 32.0 | 77.9 | 31.1 | 28.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (d)                        | (e)  | (f)                | (g) | (h)                                                | (i)  | (j)        | (k)  | (l)        | (m)  | (n)        | (o)  | (p)  | (q) |
| (4) 8.3           | 7.7  | 7.0 | DB                         | 15.0 | 35.9               | 1.7 | 0.7                                                | 13.8 | 36.4       | 12.8 | 36.8       | 11.8 | 37.2       | 10.6 | 37.7 | (4) |
| (5)               |      |     | WB                         | 13.2 | 26.9               | 1.5 | 1.1                                                | 12.1 | 27.7       | 11.2 | 28.4       | 10.4 | 29.0       | 9.3  | 29.9 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 25.0        | 27.5  | 28.4  | 28.2  | 26.9  | 25.1  | 23.4  | 22.5  | 22.5  | 22.7  | 23.1  | 23.9  | 25.6  | (6)  |      |
|                                                                        | DBStd       | 2.55        | 1.08  | 1.00  | 1.05  | 1.26  | 1.53  | 1.65  | 1.50  | 1.28  | 1.44  | 1.33  | 1.44  | 1.51  | (7)  |      |
|                                                                        | HDD10.0     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |      |
|                                                                        | HDD18.3     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (9)  |      |
|                                                                        | CDD10.0     | 5462        | 543   | 514   | 564   | 506   | 469   | 402   | 387   | 389   | 382   | 406   | 417   | 483   | (10) |      |
|                                                                        | CDD18.3     | 2421        | 285   | 281   | 306   | 256   | 211   | 152   | 128   | 131   | 132   | 148   | 167   | 224   | (11) |      |
|                                                                        | CDH23.3     | 18183       | 2591  | 2877  | 3094  | 2236  | 1479  | 797   | 535   | 551   | 649   | 776   | 974   | 1623  | (12) |      |
| (13)                                                                   | CDH26.7     | 6419        | 1052  | 1191  | 1293  | 881   | 467   | 169   | 84    | 88    | 135   | 182   | 279   | 598   | (13) |      |
| (14)                                                                   | Wind        | WSAvg       | 3.0   | 3.0   | 2.5   | 2.4   | 2.7   | 3.0   | 2.8   | 2.9   | 3.1   | 3.3   | 3.4   | 3.3   | 3.2  | (14) |
| (15) Precipitation                                                     | PrecAvg     | 46          | 4     | 14    | 24    | 5     | 1     | 1     | 1     | 0     | 0     | 1     | 1     | 2     | (15) |      |
|                                                                        | PrecMax     | 156         | 53    | 90    | 168   | 65    | 8     | 18    | 10    | 1     | 2     | 10    | 8     | 20    | (16) |      |
|                                                                        | PrecMin     | 4           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (17) |      |
|                                                                        | PrecStd     | 41          | 9     | 21    | 38    | 11    | 2     | 3     | 2     | 0     | 0     | 2     | 2     | 4     | (18) |      |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 34.2  | 35.1  | 35.1  | 33.8  | 32.5  | 31.1  | 30.1  | 30.1  | 30.9  | 30.9  | 31.9  | 33.2  | (19) |      |
|                                                                        |             | MCWB        | 25.1  | 25.9  | 25.8  | 25.1  | 24.4  | 23.7  | 23.5  | 23.2  | 23.6  | 23.6  | 24.1  | 24.6  | (20) |      |
|                                                                        | 2%          | DB          | 33.3  | 34.0  | 34.1  | 32.9  | 31.5  | 29.8  | 28.7  | 28.7  | 29.3  | 29.6  | 30.4  | 32.0  | (21) |      |
|                                                                        |             | MCWB        | 24.7  | 25.4  | 25.4  | 24.7  | 23.9  | 22.9  | 22.3  | 21.9  | 22.4  | 22.5  | 23.0  | 23.9  | (22) |      |
|                                                                        | 5%          | DB          | 32.6  | 33.1  | 33.2  | 32.1  | 30.5  | 28.6  | 27.4  | 27.5  | 28.2  | 28.7  | 29.4  | 31.1  | (23) |      |
|                                                                        |             | MCWB        | 24.3  | 25.0  | 25.0  | 24.4  | 23.6  | 22.3  | 21.5  | 21.2  | 21.6  | 21.9  | 22.4  | 23.3  | (24) |      |
|                                                                        | 10%         | DB          | 31.8  | 32.3  | 32.3  | 31.3  | 29.4  | 27.5  | 26.2  | 26.4  | 27.1  | 27.5  | 28.3  | 30.1  | (25) |      |
|                                                                        |             | MCWB        | 24.0  | 24.7  | 24.7  | 24.1  | 23.1  | 21.8  | 21.0  | 20.7  | 21.0  | 21.2  | 21.7  | 22.9  | (26) |      |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 26.2  | 27.1  | 27.2  | 26.7  | 25.3  | 24.7  | 24.6  | 24.1  | 24.6  | 24.4  | 25.0  | 25.6  | (27) |      |
|                                                                        |             | MCDB        | 32.3  | 33.0  | 32.8  | 32.0  | 30.7  | 29.8  | 29.0  | 29.4  | 29.9  | 29.9  | 30.9  | 31.8  | (28) |      |
|                                                                        | 2%          | WB          | 25.6  | 26.2  | 26.4  | 25.6  | 24.6  | 23.6  | 23.0  | 22.7  | 23.1  | 23.1  | 23.7  | 24.8  | (29) |      |
|                                                                        |             | MCDB        | 31.8  | 32.3  | 32.2  | 31.4  | 30.3  | 28.7  | 27.6  | 27.5  | 28.4  | 28.9  | 29.5  | 30.7  | (30) |      |
|                                                                        | 5%          | WB          | 25.0  | 25.7  | 25.9  | 25.0  | 24.0  | 22.7  | 22.0  | 21.7  | 22.0  | 22.2  | 22.7  | 24.0  | (31) |      |
|                                                                        |             | MCDB        | 31.3  | 31.8  | 31.7  | 31.0  | 29.7  | 27.7  | 26.6  | 26.7  | 27.5  | 28.0  | 28.6  | 29.9  | (32) |      |
|                                                                        | 10%         | WB          | 24.4  | 25.1  | 25.3  | 24.4  | 23.2  | 22.0  | 21.2  | 20.9  | 21.2  | 21.4  | 21.9  | 23.1  | (33) |      |
|                                                                        |             | MCDB        | 30.6  | 31.1  | 30.8  | 30.2  | 28.9  | 26.9  | 25.7  | 25.9  | 26.5  | 26.9  | 27.9  | 29.2  | (34) |      |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.9   | 9.3   | 9.7   | 10.1  | 9.7   | 9.2   | 9.0   | 9.8   | 10.3  | 10.5  | 10.6  | 10.5  | (35) |      |
|                                                                        |             | MCDBR       | 10.6  | 10.1  | 10.5  | 10.7  | 10.5  | 10.0  | 10.1  | 10.6  | 11.1  | 11.2  | 11.2  | 11.1  | (36) |      |
|                                                                        | 5% WB       | MCWBR       | 4.9   | 4.7   | 4.8   | 5.2   | 5.4   | 5.2   | 5.4   | 5.6   | 6.0   | 6.0   | 5.9   | 5.3   | (37) |      |
|                                                                        |             | MCWBR       | 10.1  | 9.4   | 9.6   | 10.2  | 10.2  | 9.4   | 9.5   | 10.2  | 10.8  | 11.0  | 11.1  | 10.8  | (38) |      |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.455 | 0.447 | 0.445 | 0.477 | 0.488 | 0.475 | 0.470 | 0.515 | 0.552 | 0.493 | 0.489 | 0.475 | (40) |      |
|                                                                        |             | taud        | 2.275 | 2.313 | 2.324 | 2.217 | 2.158 | 2.204 | 2.217 | 2.059 | 1.956 | 2.137 | 2.133 | 2.196 | (41) |      |
|                                                                        | Ebn at Noon | Ebn at Noon | 885   | 892   | 882   | 828   | 790   | 787   | 799   | 785   | 780   | 841   | 849   | 863   | (42) |      |
|                                                                        |             | Edh at Noon | 143   | 138   | 135   | 144   | 147   | 137   | 137   | 167   | 192   | 162   | 163   | 153   | (43) |      |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 5.70  | 5.42  | 5.70  | 5.84  | 5.29  | 4.66  | 4.77  | 5.32  | 6.07  | 6.30  | 6.38  | 6.14  | (44) |      |
|                                                                        | RadStd      | RadStd      | 0.56  | 0.34  | 0.33  | 0.25  | 0.42  | 0.55  | 0.47  | 0.42  | 0.26  | 0.24  | 0.33  | 0.35  | (45) |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page